Hint & Solution:

1. $PV = \mu RT$ ...(i)

$P\Delta V = \mu R\Delta T$ ...(ii)

Equation (ii) & (i)

$$\frac{\Delta V}{V} = \frac{\Delta T}{T}$$

$$\frac{\Delta V}{V} \times \frac{1}{\Delta T} = \frac{1}{T}$$

$$\gamma = \frac{1}{T}$$

2. α_v is a characteristic of the substance but is not strictly a constant. It depends in general on temperature as shown in figure. It is seen that α_v becomes constant only at a higher temperature.

3. The absolute minimum temperature for an ideal gas, therefore inferred extrapolating the straight line to the temprature axis, as in figure. This temperature is found to be –273.15°C and is designated as absolute zero.

4. At high altitudes, atmospheric pressure is lower so that boiling point of water is reduced.

5. Temperature versus time graph showing the changes in the state of ice on heating.

6. In figure - area of cross section is constant so T decreases linearly with x from high temperature to low temperature.

7. Heat is disordered form of energy so heat is associated with kinetic energy of random motion of molecules.

8. Sublimation curve OA represent ⇒ solid + vapour

Fusion curve AB represent ⇒ solid + liquid

Vaporisation curve AC represent ⇒ liquid + vapour

9. Convection is a mode of heat transfer by actual motion of molecules. It is possible in fluids and convection can be natural or forced.

10. $$I = \frac{ML^2}{12}$$

$$\frac{\Delta I}{I} = \frac{2\Delta L}{L} = 2\alpha\Delta T$$

$$\Delta I = 2I\alpha\Delta T$$

11. Due to anomalous behaviour, it contracts on heating between 0°C to 4°C, its density increases.

12. Due to high specific heat water warms up more slowly.

13. Due to high pressure melting point decreases. This phenomenon is called regelation.

14. (a) triple point of water is 273.16K

(b) Steam at 100°C carries more heat than water at 100°C due to latent heat of vaporisation

(c) Ethyl alcohol $\alpha = 110 \times 10^{-5}\ K^{-1}$ and $\alpha_{Hg} = 18.2 \times 10^{-5}$

(d) it will contract due to contraction of air inside.

15. In conduction and convection medium must be required and in radiation heat is transfered (or radiated) by electromagnetic waves.

16. Air at the earth's surface near equator is hotter while the air at the poles is colder. So the convection current would be set up which causes the air at the equatorial surface, rising and moving out towards the poles.

17. According to stefan's-boltzman law, energy of thermal radiation emitted per unit time by a black body $\varepsilon = \sigma AT^4$. But bodies which are not black body, emits less radiation. It is however proportional to T^4.

18. Newton's law of cooling is special case of stefan's boltzman law.

19. Thermal conductivity depends on material of object. Solids are better conductors than liquids and liquids are better conductor than gases.

20. Aluminium having higher $\propto$ and will expand more.

21. angular speed decreases due to increase in moment of inertia.

$$I_1\omega_1 = I_2\omega_2$$

22. Density of water will be maximum at 4ºC. Therefore buoyancy will also be maximum at 4ºC $F_B = \rho_w Vg$.

23. Length of rod increases due to thermal expansion. However centre of mass of bob remains at centre of bob due to its uniform expansion.

24. $\frac{dT}{dt} = \frac{-\sigma eA}{ms}\left[T^4 - T_s^4\right]$

$\therefore \frac{dT}{dt} \propto A$

Plate having maximum area will cool fastest and sphere having lowest area will cool slowest.

25. Y and Z may have equal temperature but different from that of X.

26. Water and ice remain in thermal equilibrium at 0°C until phase change gets completed.

27. Invar has extremely small temperature co-efficient of expansion

$\therefore$ The length of pendulum remains almost constant, giving correct time, in summar & winter.

28. Black materials are good absorbers of heat.

Therefore they maximize the utilization of heat.

29. Due to presence of medium (air), heat will be lost through conduction and convection. Radiation will take place irrespective of medium at all temperatures.